

Requirements Table

Centralized Audit Trail Compliance Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall parse structured JSON audit logs, identify and mask PII fields such as Social Security Numbers and credit-card numbers before persisting records to storage.	High	Pass: The stored audit log contains masked PII and a SHA-256 integrity hash. Fail: Any configured PII such as a credit-card number is stored in clear text.	Prevents sensitive personal information from being exposed in stored audit records.	
FR-002	Functional	The system shall ingest structured JSON audit logs generated by enterprise applications.	High	Pass: A valid structured JSON audit event submitted by an enterprise application is successfully accepted for processing. Fail: A valid JSON audit event is rejected without a processing error.	Centralized audit logging requires the system to collect events generated by enterprise applications.	
FR-003	Functional	The system shall allow authorized auditors to search and filter historical audit records using available audit fields such as time range, application, user, and event type.	High	Pass: When one or more filters are applied, every returned audit record satisfies the selected filter conditions. Fail: A returned record does not satisfy the requested filters.	Compliance and security investigations require specific historical events to be located efficiently.	
FR-004	Functional	The system shall generate an immutable audit export containing selected audit records and their associated integrity information.	High	Pass: An export request produces a file containing all selected audit records and their integrity information. Fail: Selected records or required integrity information are missing from the generated export.	Audit evidence must be preserved and provided in a trustworthy form for compliance reviews.	
FR-005	Functional	The system shall verify the cryptographic integrity of an audit export and indicate whether the export has been modified after generation.	High	Pass: An unchanged export is reported as valid, while an intentionally modified export is reported as invalid. Fail: A modified export is reported as valid.	Auditors need to determine whether exported audit evidence has been altered.	
NFR-001	Nonfunctional - Performance	The system shall support complex filtered queries across 10 million historical audit records in under 1 second.	High	Pass: Benchmark tests confirm that the query completes in less than 1 second across 10 million records under the defined test load. Fail: The query requires 1 second or more.	Fast retrieval is necessary when investigating large volumes of enterprise audit data.	
NFR-002	Nonfunctional - Security / Integrity	The system shall ensure that unauthorized modification of audit exports is detectable through cryptographic integrity verification.	High	Pass: Changing at least one value in a generated audit export causes its integrity verification to fail. Fail: A modified export successfully passes integrity verification.	Protects the trustworthiness of audit evidence and helps reveal tampering.	